

Proofs

Theorem 1.1. (The Division Theorem for $\mathbb{Z}$):
 $a, b \in \mathbb{Z}$ with $b > 0 \exists! q, r \in \mathbb{Z}$ with $0 \leq r < b$ s.t.
 $a = bq + r$.

Proof. (Existence): Let
 $R = \{a - bm : m \in \mathbb{Z} \text{ \& } a - bm \geq 0\}$, then as: if
 $a \geq 0$, $a - b \cdot 0 = a \geq 0$ so $a \in R$; if $a < 0$,
 $a - b \cdot a = a(1 - b) \geq 0$ so $a - b \cdot a \in R$, $R \neq \emptyset$.
Thus, take $r = \min(R)$ & so $0 \leq r = a - bm \in R$
for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by
construction $a = bq + r$. Note, $r < b$ as were such
not true then $r > a - b(q + 1) = r - b \in R$ but
this contradicts $r = \min(R)$, so we have
 $0 \leq r < b$. (*Uniqueness*): Suppose $q, q', r, r' \in \mathbb{Z}$
with $0 \leq r', r < b$ & $a = bq + r = bq' + r'$, also
WLOG take $r' \leq r$. So with $\frac{r-r'}{b} \in (-1, 1)$, &
 $\frac{r-r'}{b} = \frac{a-bq-(a-bq')}{b} = q' - q \in \mathbb{Z}$, we have
 $\frac{r-r'}{b} = 0$ i.e. $r = r'$. So
 $q = \frac{a-r}{b} = \frac{a-r'}{b} = q'$. $\square$

Euclidian Algorithm:

Let $a, b \in \mathbb{Z}^+$, apply Th^m 1.1, to give:

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \\ r_{n-1} &= r_nq_{n+1}, \end{aligned}$$

Here, $\gcd(a, b) = r_n$. To prove this relation
 $G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2})$, where
 $G(a, b) = \{n \in \mathbb{Z} : n|a \text{ \& } n|b\}$.

Reverse Euclid:

Given a completed algorithm we can obtain
 $n, m \in \mathbb{Z}$ s.t. $na + mb = \gcd(a, b)$ i.e. taking the
end of a Euclidian algoirthm:

$$\begin{aligned} &\vdots \\ r_{n-3} &= r_{n-2}q_{n-1} + r_{n-1} \\ r_{n-2} &= r_{n-1}q_n + r_n \\ r_{n-1} &= r_nq_{n+1} \end{aligned} \quad \begin{aligned} &(1) \\ &(2) \\ &(3) \end{aligned}$$

$$\begin{aligned} (3) &\implies r_n = r_{n-2} - r_{n-1}q_n & (4) \\ (2) &\implies r_{n-1} = r_{n-3} - r_{n-2}q_{n-1} & (5) \\ (4) \text{ \& } (5) &\implies r_n = r_{n-2} - & \\ &\quad (r_{n-3} - r_{n-2}q_{n-1})q_n & \\ &\vdots & \end{aligned}$$

we continue this ladder up, simplifying each time,
(Notes: $r_n = \gcd(a, b)$)

Theorem 1.9. Every integer $n \geq 2$ can be
written as a product of primes.

Proof. (By strong induction):

(Base case, $n = 2$): Trivially true, $n = 2$ a
product of primes.

(Inductive Hypothesis): Suppose for
 $n \in 2, 3, \dots, k$ $n = p_{n,1}p_{n,2} \dots p_{n,m_n}$.

Then for $n = k + 1$, if $k + 1$ is prime we are done,
if not then $k + 1 = ab$ where $a, b \in \{2, 3, \dots, k\}$,
thus $a = p_{a,1}p_{a,2} \dots p_{a,m_a}$ &
 $b = p_{b,1}p_{b,2} \dots p_{b,m_b}$ so:
 $k + 1 = p_{a,1}p_{a,2} \dots p_{a,m_a}p_{b,1}p_{b,2} \dots p_{b,m_b}$.
So $\forall n \in \mathbb{N}$ with $n \geq 2$, we are done. $\square$

Theorem 1.10. (Euclid's property of primes)
 $\forall p \in \text{Primes} \text{ \& } \forall a, b \in \mathbb{Z}. p|ab \implies p|a \text{ or } p|b$.

Proof. (Case $p|a$): We are done.

(Case $p \nmid a$): Then using the Euclidian algorithm
we obtain: $1 = na - mp$ with $n, m \in \mathbb{Z}$. So
 $p|b \iff p|(na - mp)b \iff p|n(ab) - mbp \iff$
 $p|ab$. $\square$

the chain of implications uses a proposition in the notes
which states: $p|a \text{ \& } p|b \implies p|a + b \text{ \& } p|ab$.

Theorem 1.12. (Fundamental Theorem of
Arithmetic) $\forall n \in \mathbb{N}$ with $n \geq 2$,
 $\exists! p_1, p_2, \dots, p_r \in \text{Primes}$ up to ordering, s.t.:
 $n = p_1p_2 \dots p_r$.

Proof. The existence is from Th^m 1.9. For
uniqueness, suppose $p_1p_2 \dots p_r = q_1q_2 \dots q_s$, then
cancel common primes, suppose the result is not
1 i.e. we are left with $p_1p_2 \dots p_n = q_1q_2 \dots q_m$
with $p_i \neq q_j, \forall i \in \{1, \dots, n\} \text{ \& } \forall j \in \{1, \dots, m\}$,
after reindexing. Then $p_1|q_1q_2 \dots q_m$ so
 $\exists j \in \{1, \dots, m\} p_1|q_j$, but q_j is prime so $p_1 = q_j$,
 $\cancel{p_1}$. $\square$

Lemma 1.16. $\forall a, b \in R: 0a = a0 = 0$;

$a(-b) = (-a)b = -(ab)$; & $(-a)(-b) = ab$

Proof. $0a + 0 = 0a = (0 + 0)a = 0a + 0a$ so
 $0a = 0$.

$ab + a(-b) = a(b - b) = a0 = 0 = (a - a)b =$
 $ab + (-a)b$ so by the Groups & Geo^m we have
 $(-ab)$ is the unique element s.t. $ab - (ab) = 0$,
 $a(-b) = -(ab) = (-a)b$.

$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. $\square$

Lemma 1.18. (Subring Test) Let R be a ring &
 $S \subseteq R$. Then S is a subring of R iff the following
hold, $\forall r, s \in S$:

1. $1_R \in S$,
2. $r + s, rs \in S$, &
3. $-r \in S$.

Proof. By 1 & 3 we have $-1_R \in S$ so

$0_R = 1_R - 1_R \in S$ by 2. So $(S, +)$ is a group by
the subgroup test with $(R, +)$. Commutativity of
 $+$, associativity of $+$ & $\times$, & distributivity of $\times$
over $+$ are all inherited from R . Note, as they
shair operations they shair the identities of the
operations, so as, $0_R \neq 1_R$ in R we have
 $0_S = 0_R \neq 1_R = 1_S$ in S , so (R1)-(R4) are
satisfied.

Lemma 2.2. Suppose $\text{Char}(R) = n > 0$, then:

- (i) $n \cdot r = 0, \forall r \in R$; & (ii) if $m \in \mathbb{Z}^+$ then
 $m \cdot 1 = 0$ iff $n|m$.

Proof. (i):

$$\begin{aligned} n \cdot r &= \underbrace{r + r + \dots + r}_{n \text{ times}} \\ &= r \underbrace{(1 + 1 + \dots + 1)}_{n \text{ times}} = r0 = 0 \end{aligned}$$

- (ii): ($\implies$): Let $m \cdot 1 = 0$ with $m \in \mathbb{Z}^+$, applying
the division Th_m with $m = nq + r$ where $q, r \in \mathbb{Z}$
& $0 \leq r < n$ so $0 = m \cdot 1 = (nq + r) \cdot 1 =$
 $q(n \cdot 1) + r \cdot 1 = q \cdot 0 + r \cdot 1 = r \cdot 1$, so $r = 0$ as
otherwise $n > r \in \mathbb{Z}^+ \text{ \& } r \cdot 1 = 0$, a $\cancel{r}$ of n 's def^m.
(impliedby): Let $n|m$ i.e. $\exists q \in \mathbb{Z}$ s.t. $m = qn$, so
 $m \cdot 1 = qn \cdot 1 = q(n \cdot 1) = q \cdot 0 = 0$. $\square$

Lemma 2.5. If S is a subring of R & R is a
domain then S is a domain.

Proof. Let $r, s \in S$ with $rs = 0$, thus $rs = 0$ in R ,
hence $r = 0$ or $s = 0$. $\square$

Proposition 2.7. Suppose R is a domain. Then
 $R[X]$ is a domain.

Proof. We use the contrapositive of the definition
of a domain [i.e.: R a domain iff
 $\forall 0 \neq r, s \in R, rs \neq 0$]. Let $0 \neq f, g \in R[X]$, taking
the leading coefficients of f & g as $a \neq 0$ & $b \neq 0$,
respectively, we the leading coefficient of fg as
 $ab \neq 0$, hence $fg \neq 0$. $\square$

Note: This immediately (by induction on n) yields
 $R[X_1, \dots, X_n]$ & domain if R is a domain.

Lemma 2.11. If R is a ring & $r \in R$ has both a
left & right inverse then these equal & unique.

Proof. (Equal): Choose, $rs = 1 = tr$, then:
 $s = 1s = trs = t1 = t$.

(Unique right inverse): Let $rs = 1 = rs'$ then
 $s = 1s = trs = t1 = trs' = 1s' = s'$, where t is
choosen s.t. $tr = 1$.

(Unique left inverse): Let $tr = 1 = t'r$ then
 $t = t1 = trs = 1s = t'rs = t'1 = t'$, where s is
choosen s.t. $rs = 1$ $\square$

Proposition 2.13. Every division ring is a
domain (& subsequently every field is an integral
domain).

Proof. Let R be a division ring & suppose
 $r, s \in R$ & $rs = 0$. Then assume $r \neq 0$ so r^{-1}
exists, then $s = r^{-1}rs = r^{-1}0 = 0$. Then given
the immediate other case of $r = 0$ in all cases
either $r = 0$ or $s = 0$. Hence R is a domain. $\square$

Lemma 2.12. The following conditions for an
integer $n \geq 2$ are equivalent:

- (i) $\mathbb{Z}_n$ is an integral domain;
- (ii) $\mathbb{Z}_n$ is a field;
- (iii) n is prime.

Proof. (i) $\implies$ (ii): Suppose $\mathbb{Z}_n$ is a domain & let
 $[m]_n \in \mathbb{Z}_n$ & $[m]_n \neq 0$. Consider the list
 $[0]_n[m]_n, [1]_n[m]_n, \dots, [n-1]_n[m]_n$, each element
in the list is distinct as, suppose otherwise then
 $[i]_n[m]_n = [j]_n[m]_n$ for some distinct
 $i, j \in \{0, 1, \dots, n-1\}$, then $([i]_n - [j]_n)[m]_n = 0$
but $\mathbb{Z}_n$ is suppose to be a domain & $[m]_n \neq 0$ so
 $[i]_n - [j]_n = [0]_n$ i.e. $[i]_n = [j]_n$, $\cancel{i}$. Then as there
are n elements of the list each in $\mathbb{Z}_n$, then
 $\{[0m]_n, [1m]_n, \dots, [(n-1)m]_n\} = \mathbb{Z}_n$. So some
multiple of $[m]_n$ say $[r]_n \in \mathbb{Z}_n$ is s.t.
 $[m]_n[r]_n = [r]_n[m]_n = [1]_n$.

(ii) $\implies$ (iii): Suppose $\mathbb{Z}_n$ is a field then suppose
 $m, l \in \mathbb{N}$ are s.t. $ml = n$, so $0 < m \leq l \leq n$. Then
 $[m]_n[l]_n = [ml]_n = [n]_n = [0]_n$ so as fields are
domains, $[m]_n = [0]_n = [n]_n$ or $[l]_n = [0]_n = [n]_n$
& so, by the equivalence class definition, $m = n$
or $l = n$ as neither are 0. So if $m = n$, $l = 1$ & if
 $l = n$, $m = 1$, so the only factors of n are 1 & n in
 $\mathbb{N}$, i.e. n is prime.

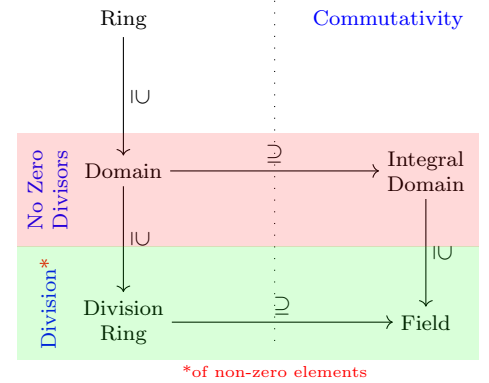
(iii) $\implies$ (i): Suppose n is prime & let
 $[m]_n, [l]_n \in \mathbb{Z}_n$ s.t. $[m]_n[l]_n = 0$ i.e. $[ml]_n = [0]_n$
defintionally $n | ml$ so by Th^m 1.10 $n | m$ or $n | l$
i.e. $[m]_n = [0]_n$ or $[l]_n = [0]_n$. So with $\mathbb{Z}_n$'s
commutativity, $\mathbb{Z}_n$ is an integral domain. $\square$

Proposition T1. In a domain R if $ab = ac$ or
 $ba = ca$ for $a, b, c \in R$ & $a \neq 0$ then $b = c$.

Proof. If $ab = ac$, then $a(b - c) = 0$. If $ba = ca$,
then $(b - c)a = 0$. Since $a \neq 0$ and R is a domain,
 $b - c = 0$. $\square$

Lemma 2.15. In a ring R , the set of units R^*
forms a group under multiplication.

Proof. (G0): $1 \cdot 1 = 1$ so $1 \in R^*$ i.e. $R^* \neq \emptyset$;
(G1): given (R2) we have associativity & given
 $r, s \in R^*$, $(rs)(s^{-1}r^{-1}) = r(ss^{-1})r^{-1} = 1$ &
 $(s^{-1}r^{-1})(rs) = s^{-1}(r^{-1}r)s = 1$, so $rs \in R^*$ as
required; (G2): (R4); & (G3): Let $r \in R^*$ so r is
a unit so $r^{-1} \in R^*$ exists, which itself has inverse
 $(r^{-1})^{-1} = r \in R$, so $r^{-1} \in R^*$, as required. $\square$



Lemma 3.3. Suppose $\varphi : R \rightarrow S$ is an
isomorphism. Then: (i) $\varphi(1_R) = 1_S$; (ii)
 $\varphi(0_R) = 0_S$; (iii) $\varphi(-r) = -\varphi(r)$, $\forall r \in R$; (iv)
 $r \in R$ is invertible iff $\varphi(r) \in S$ is invertible and, in
that case, $(\varphi(r))^{-1} = \varphi(r^{-1})$; (v) $r \in R$ is
nilpotent iff $\varphi(r)$ is nilpotent and, in that case,
they have the same index of nilpotence.

Proof. (i): Let $s \in S$. Since φ is bijective,
 $s = \varphi(r)$ for some $r \in R$. Then
 $\varphi(1_R)s = \varphi(1_R)\varphi(r) = \varphi(r) = s$ and similarly
 $s\varphi(1_R) = s$, hence $\varphi(1_R) = 1_S$.

(ii): $\varphi(0_R) + 1_S = \varphi(0_R) + \varphi(1_R) = \varphi(1_R) = 1_S$,
so $\varphi(0_R) = 0_S$.

(iii): $\varphi(-r) + \varphi(r) = \varphi(0_R) = 0_S$, so
 $\varphi(-r) = -\varphi(r)$.

(iv): If r is invertible then
 $\varphi(r^{-1})\varphi(r) = \varphi(r^{-1}r) = 1_S$ and
 $\varphi(r)\varphi(r^{-1}) = \varphi(rr^{-1}) = 1_S$, so $\varphi(r)$ is invertible
with $(\varphi(r))^{-1} = \varphi(r^{-1})$. Conversely, apply this
to $\varphi^{-1} : S \rightarrow R$.

(v): Since $(\varphi(r))^n = \varphi(r^n)$, $r^n = 0_R$ iff
 $(\varphi(r))^n = 0_S$. Thus the least such n is the
same. $\square$

Definitions

Definition 1.4. $\forall a, b \in \mathbb{Z}$,

$\gcd(a, b) = \max\{n \in \mathbb{Z} : n|a \text{ \& } n|b\}$, noting
 $\gcd(0, 0) = 0$ & if $\gcd(a, b) = 1$ then a & b are coprime.

Definition 1.6. $\forall a, b, n \in \mathbb{Z}$ with $n \geq 2$. a is **congruent to b modulo n** iff $n|a - b$, written $a \equiv b \pmod n$. Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod n\} = \{b \in \mathbb{Z} : n \mid (b - a)\} = \{a + kn : k \in \mathbb{Z}\}$
 We thus have:

$$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, [2]_n, \dots, [n-1]_n\}$$

Note: $b \equiv a \pmod n$ iff $b = a + kn$ iff

$$[a]_n = \{a + nk : k \in \mathbb{Z}\} \text{ iff } [a]_n = [a + kn]_n$$

Definition 1.14. A **ring** is a set R with two binary operations $+$ & $\times$, defined on R , satisfying the following:

- (R1) $(R, +)$ is an abelian group with identity 0.
- (R2) $\times$ is associative.
- (R3) $\times$ distributes over $+$ i.e.:

$$a(b+c) = ab+ac, \text{ \& } (a+b)c = ac+bc, \forall a, b, c \in R$$

(R4) The multiplicative identity $1 \in R$ & $1 \neq 0$

Example. With $n \geq 2$, $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only two-sided ideals of $M_n(F)$, F a field, are $M_n(F)$ & $\{0\}$ but this isn't a division ring hence even relaxing commutative in Prop 3.22 doesn't yield the same prop with field $\rightarrow$ division ring.

Example. For X a set $\mathcal{P}(X)$ with

$$A + B = (A \cup B) \setminus (A \cap B), \quad AB = A \cap B$$

& $0 = \emptyset$, $-A = A$, $A, B \in \mathcal{P}(X)$.

Definition 1.17. Let R be a ring & $S \subseteq R$.

Then S is a **subring** of R if S is a ring w.r.t. the same addition & multiplication as R & contains the multiplicative identity of R , 1_R .

Definition 1.21. let R be a ring. The **ring of polynomials**

$R[X] = \{\sum_{i \geq 0}^n a_i X^i \text{ a finite sum} : a_i \in R\}$ with addition & multiplication defined as:

$$\sum_{i \geq 0}^n a_i X^i + \sum_{i \geq 0}^m b_i X^i = \sum_{i \geq 0}^{\max(n, m)} (a_i + b_i) X^i$$

$$\& \left(\sum_{i \geq 0}^n a_i X^i \right) \left(\sum_{i \geq 0}^m b_i X^i \right) = \sum_{i \geq 0}^{n+m} \left(\sum_{j=0}^i a_j b_{i-j} \right) X^i$$

$0 = \sum_{i \geq 0} 0X^i$ & $1 = 1X^0 + \sum_{i \geq 1} 0X^i$. Then the

degree of a polynomial is the greatest $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \neq 0$ or if no such i exist $\deg(0) = -\infty$

More generally, we have:

$$R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$$

$\forall f, g \in R[X], \deg(f+g) \leq \max(\deg(f), \deg(g))$ &
 $\deg(fg) \leq \deg(f) + \deg(g)$.

Definition 1.24. If R_1 & R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with the $+$ & $\times$ defined element wise is a ring.

Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$,
 $n \cdot a = \underbrace{a + a + \dots + a}_{n \text{ times}}$; for $n = 0$, $n \cdot a = 0 \cdot a = 0$;

& for $n \in \mathbb{Z}^-$, $n \cdot a = (-n) \cdot (-a)$.

The **Characteristic**, $\text{char}(R)$, of a ring R is the least positive integer n s.t. $n \cdot 1 = 0$, or if no exist $\text{char}(R) = 0$.

Note: $(n+m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a+b) = n \cdot a + n \cdot b$, & $n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.

Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.

A ring R is a domain if, $\forall r, s \in R$:

$$rs = 0 \implies r = 0 \text{ or } s = 0.$$

i.e. a ring with no zero divisors is a domain & vice versa.

Definition 2.9. A **division ring** is a ring where every non-zero element has a right & left inverse, i.e.: $\forall r \in R \setminus \{0\}, \exists s, t \in R$ s.t. $rs = 1 = tr$ with s, t being the right & left inverses, respectively.

Note: this immediately collapse, due to Lemma 2.11, to just there exists an inverse as $s = t$.

An **invertible** element r is called a **unit**.

Example. Division rings are not all fields e.g. the quaternions $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where $i^2 = j^2 = k^2 = -1$ & $ij = k$, $jk = i$, & $ki = j$, this is non-commutative as $-ij = -k = jk = ji$ so $ij \neq ji$ and this is a division ring as for $z = a + bi + cj + dk$
 $z^{-1} = \frac{a-bi-cj-dk}{a^2+b^2+c^2+d^2}$, taking $z \neq 0$.

Definition 2.16. An element r of a ring R is **nilpotent** if $\exists n \in \mathbb{Z}^+$ s.t. $r^n = 0$, and the least such n is the **index of nilpotence** of r .

An element $r \in R$ is **idempotent** if $r^2 = r$.

Note: In any ring R , 0_R is nilpotent with index 1 and 0_R and 1_R are idempotent.

Definition 3.1. If R & S are rings then an **isomorphism** from R to S is a bijection

$\varphi : R \rightarrow S$ s.t. $\forall r, r' \in R$:

$$\varphi(r + r') = \varphi(r) + \varphi(r') \quad \& \quad \varphi(rr') = \varphi(r)\varphi(r').$$

If φ is an isomorphism from R to S then R & S are **isomorphic**, written $R \cong S$.